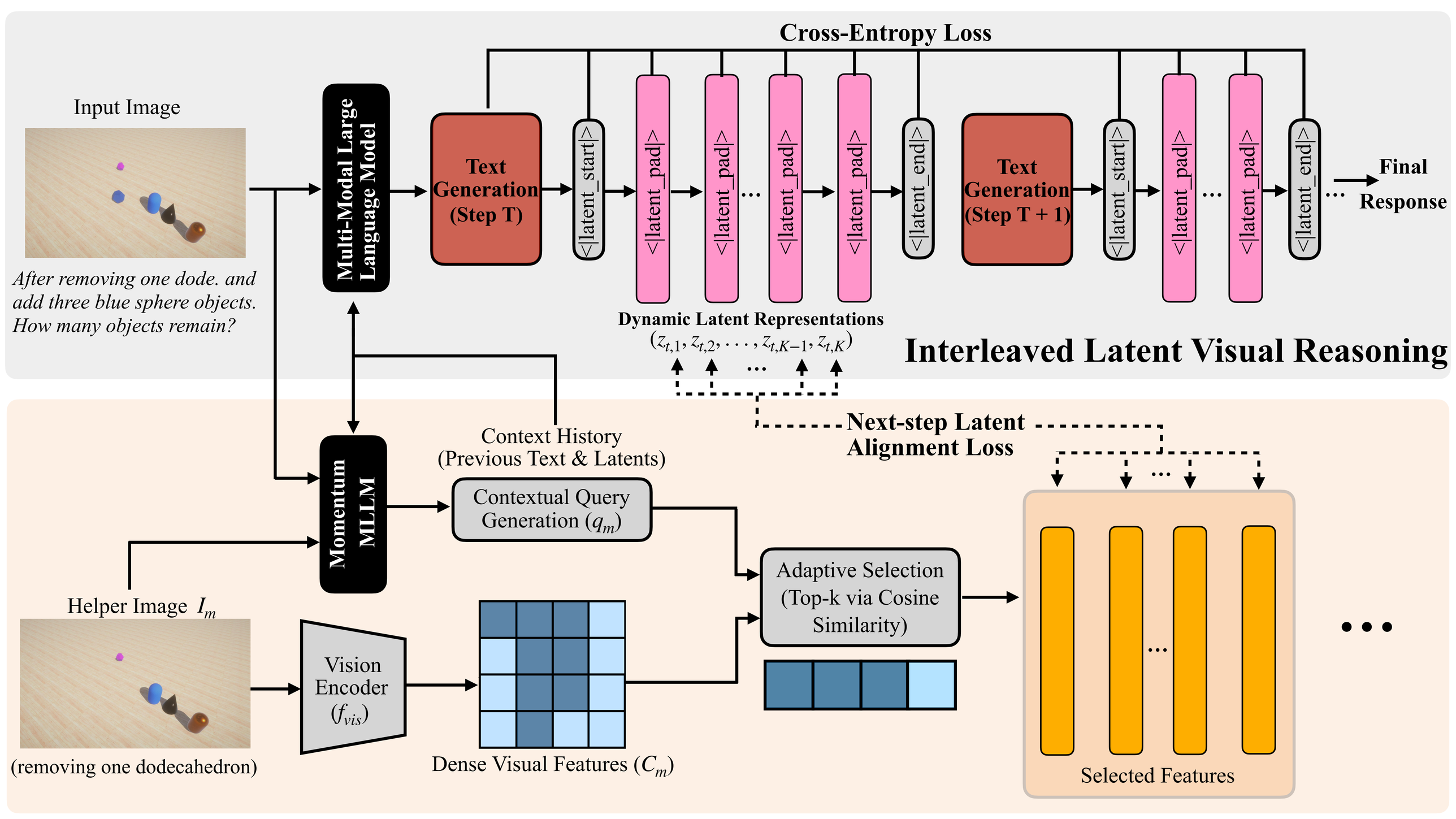
Cross-Entropy Loss
Input Image
After removing one dode. and add three blue sphere objects. How many objects remain?
Multi-Modal Large Language Model
Text Generation (Step T)
<|latent_start|>
<|latent_pad|>
<|latent_pad|>
<|latent_pad|>
<|latent_pad|>
<|latent_end|>
Text Generation (Step T + 1)
<|latent_start|>
<|latent_pad|>
<|latent_pad|>
<|latent_end|>
Final Response
Dynamic Latent Representations
$(z_{t,1}, z_{t,2}, \ldots, z_{t,K-1}, z_{t,K})$
Interleaved Latent Visual Reasoning
Next-step Latent Alignment Loss
Context History (Previous Text & Latents)
Momentum MLLM
Contextual Query Generation (q_m)
Helper Image I_m
(removing one dodecahedron)
Vision Encoder (f_{vis})
Dense Visual Features (C_m)
Adaptive Selection (Top-k via Cosine Similarity)
Selected Features